Drift Analysis of MOEA/D under Prior Noise without Re-evaluation

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Abstract

We study MOEA/D on OneMinMax with Tchebycheff decomposition, neighbourhood size one, and noisy evaluations that are stored without re-evaluation. For one-bit noise, either one-bit mutation or standard bit mutation finds the exact solution of every subproblem in $O(n^2 \log n)$ expected evaluations. For standard bit mutation with activated bitwise noise, we prove $O(n^2 \log n[1 + np \log(e/p)])$ expected evaluations. Both bounds hold for $p \leq 1/(16e(n+1))$. The second is $O(n^2 \log^2 n)$ throughout this range and $O(n^2 \log n)$ when $p = O(1/(n \log n))$. The proofs separate progress in stored values from correction of the true solutions. An expected-repair-time potential handles the second task for all four mutation–noise combinations. However, one-bit mutation with bitwise noise behaves differently: at $p = n^{-4}$, its expected exact-covering time is $\Omega(n^{5/2})$ evaluations along an infinite subsequence of dimensions. Thus the same $O(n^2 \log n)$ exact-covering bound cannot hold uniformly for all four combinations.

1 Introduction

A noisy evaluation can make a poor solution look good. If an evolutionary algorithm stores that evaluation, the error can influence selection for many subsequent iterations. Runtime analysis must therefore account for both optimization progress and the time needed to correct misleading stored values.

We study this issue for MOEA/D on OneMinMax. There are $n+1$ subproblems, one for each possible Hamming weight. Each subproblem keeps its own solution and stored evaluation. The algorithm updates every subproblem once per sweep. Throughout the paper, the neighbourhood size is $T = 1$: an offspring can replace only the solution of the subproblem that generated it.

The main proof idea is to separate two kinds of progress. First, the stored evaluations must reach their target values. Second, the true solutions must match those values simultaneously. The second task admits one common potential for all four mutation–noise combinations: sum the expected time needed to repair each subproblem. The first task depends more strongly on the operators. One-bit noise creates only a one-step discrepancy between true and observed weight. Bitwise noise can create larger discrepancies, so their expected repair cost must be included explicitly.

Related work. Dinot, Doerr, Hennebelle, and Will [2] analyzed noise robustness of SEMO on OneMinMax. Antipov and Doerr [1] showed that omitting re-evaluation can substantially improve robustness on LeadingOnes. Our setting uses decomposition-based selection. The choice of decomposition also matters: Zhang et al. [4] analyze noise-free MOEA/D with weighted sums, whereas we use Tchebycheff scalarization.

Diversity-based parent selection [5] and dynamic population sizes [6] can accelerate multi-objective search by concentrating evaluations on useful parents. Here the cyclic schedule and all $n + 1$ subproblems remain fixed. These algorithmic speed-ups provide motivation, but are not assumptions of our analysis. We use additive drift, multiplicative drift, and fitness-level arguments, following the runtime-analysis approach exemplified by [3]. All potential estimates needed for the proofs are derived below.

2 Algorithm, results, and proof structure

2.1 The algorithm

Let $n \geq 2$. On a bit string $x \in \{0,1\}^n$, OneMinMax maximizes the two objectives

$$f(x) = (|x|_1, n - |x|_1).$$

Its Pareto front contains one objective vector for each weight $0, \ldots, n$. For each subproblem $k \in \{0, \ldots, n\}$, the algorithm stores a string x_k and one observed weight b_k. We write

$$j_k = |x_k|_1 \quad \text{for its true weight,} \qquad b_k \quad \text{for its stored observed weight.}$$

We also call subproblem k a *coordinate* of the population. Its state is *natural* when $j_k = b_k$. This means equality of weights, not necessarily equality of the true and noisy strings.

The reference point $z = (z_1, z_2)$ is the componentwise maximum of every objective vector observed so far, including rejected offspring. Subproblem k minimizes

$$g_k(b, z) = \max\left\{\frac{k}{n}|z_1 - b|, \frac{n-k}{n}|z_2 - (n-b)|\right\}.$$

Each sweep processes $k = 0, 1, \ldots, n$ in this order. The update of subproblem k consists of four steps:

1. Mutate x_k to obtain an offspring y.
2. Evaluate a noisy copy of y, obtaining its observed weight c.
3. Update the reference to $z \leftarrow (\max\{z_1, c\}, \max\{z_2, n - c\})$.
4. If $g_k(c, z) \leq g_k(b_k, z)$, replace (x_k, b_k) by (y, c).

Both scalar values in the last step use the updated reference. Ties are accepted. Noise changes the evaluation only: the retained offspring is y, not its noisy copy. Parents are not re-evaluated, and there is no exchange between subproblems.

Mutation and noise. One-bit mutation flips one uniformly chosen bit. Standard bit mutation flips each bit independently with probability $1/n$. For both noise models, an evaluation is clean with probability $1 - p$. With probability p, noise is activated and acts as follows:

- One-bit noise flips one uniformly chosen bit of the evaluated string.
- Activated bitwise noise flips each bit independently with probability $1/n$.

Both objectives are evaluated on the same noisy copy. All mutation and noise choices are independent across updates. Activated bitwise noise differs from flipping each bit with probability p/n at every evaluation: activation is a single event shared by all bits.

Initialization. The original algorithm samples $x_0, \ldots, x_n$ independently and uniformly, evaluates each once, and sets z to the componentwise maximum of those observations. Our upper bounds also allow arbitrary true initial strings, provided they are evaluated independently by the stated noise model. Under one-bit noise we prove a stronger statement from every *admissible* initial state: $|j_k - b_k| \leq 1$, the reference belongs to $[0, n]^2$ and dominates all stored objective vectors, and $j_k = b_k$ if $p = 0$.

Success criterion and time units. *Exact covering* means $j_k = k$ for every k simultaneously. Let T_{exact} be the first sweep boundary at which this happens. Exact covering implies Pareto-front coverage. The converse need not hold, because front coverage permits the weights to be assigned to different subproblems. One sweep costs $n+1$ evaluations, and initialization costs another $n+1$. We state local bounds in updates of one coordinate and global bounds in sweeps, converting to evaluations at the end. Throughout, log is the natural logarithm and $H_m = \sum_{r=1}^{m} 1/r$, with $H_0 = 0$. The symbol $\mathcal{F}_t$ denotes the history before the transition indexed by t.

2.2 Results

Theorem 1 (Upper bounds)**.** *For $0 \le p \le 1/(16e(n+1))$, the expected evaluation cost of exact covering is*

$$\begin{cases} O(n^2 \log n), & \text{one-bit noise, either mutation operator,} \\ O\big(n^2 \log n[1 + np \log(e/p)]\big), & \text{activated bitwise noise, standard bit mutation.} \end{cases}$$

At $p = 0$, the second bound is understood as $O(n^2 \log n)$. The first statement holds from every admissible initial state. The second holds for arbitrary true initial strings evaluated independently by the stated noise model, with the reference initialized from their observations.

The bitwise–bitwise bound is $O(n^2 \log^2 n)$ throughout the stated range. It improves to $O(n^2 \log n)$ when $p = O(1/(n \log n))$. For one-bit mutation with activated bitwise noise, theorem 17 instead proves an $\Omega(n^{5/2})$ exact-covering lower bound at $p = n^{-4}$ along an infinite subsequence of dimensions. That lower bound concerns exact covering, not front coverage with arbitrary assignment.

2.3 Why the proof has two phases

When $z = (n, n)$, the scalar objective becomes

$$f_k(b) := g_k(b, (n, n)) = \max\{k(n-b)/n, (n-k)b/n\}.$$

It strictly decreases up to $b = k$ and strictly increases afterwards. Thus its unique minimizing weight is k. Once a coordinate stores $b_k = k$ under this reference, it accepts only observations of weight k. We call this observation *pinned*. A coordinate is *active* while $b_k \neq k$. These descriptions concern the stored value. A coordinate is *truly correct* when $j_k = k$, and *natural* when $j_k = b_k$.

We use two stopping times, both measured at sweep boundaries:

1. U is the first boundary with the ideal reference $z = (n, n)$.
2. S is the first boundary with $z = (n, n)$ and $b_k = k$ for every k.

Phase I ends at S. It consists of reaching the ideal reference and then reaching all stored targets. *Phase II* starts at S and ends when every true weight also equals its target. The stored targets stay fixed in Phase II, but the true weights can still change.

The proof follows these tasks. Section 3 bounds Phase I under one-bit noise using a three-valued potential on each fitness level. Section 4 gives the common Phase-II potential and completes the one-bit-noise bounds. Sections 5 and 6 handle bitwise mutation with bitwise noise: a harmonic potential counts natural updates, while an expected-repair-time potential controls the remaining updates. Section 7 explains the obstruction for one-bit mutation with bitwise noise, with the detailed probability estimates in appendix A.

3 One-bit noise: a three-valued level potential

We first analyze the time needed to make the *stored* weights correct. Throughout this section, noise is one-bit and the mutation operator is either one-bit mutation or standard bit mutation. One-bit noise gives the useful invariant

$$|j_k - b_k| \le 1.$$

Thus the true and stored weights always differ by at most one.

Fix a coordinate k and, initially, keep the reference at (n, n). For each nonoptimal value c of its scalar function f_k, define

$$I_c = \{b \in \{0, \dots, n\} : f_k(b) \le c\}, \qquad E_c = \{b \in \{0, \dots, n\} : f_k(b) = c\},$$

$$d(c) = \min_{b \in E_c} |b - k|.$$

The set I_c contains the observed weights that would be accepted from level c. It is an integer interval containing k. Since f_k decreases strictly up to k and increases strictly after k, the level set E_c contains at most two weights, each at a boundary of I_c. In particular, $d(c) \ge 1$. For each fixed distance d, at most two levels satisfy $d(c) = d$: one must contain $k - d$ or $k + d$.

Lemma 2 (Local progress)**.** *Suppose $p \le 1/2$. Fix a nonoptimal scalar level c and put $s = d(c)/(2en)$. In one coordinate update:*

1. *from a natural state on level c, the probability of reaching a strictly smaller level is at least s;*
2. *from a non-natural state on level c, the probability of either becoming natural on that level or reaching a strictly smaller level is at least s;*
3. *from a natural state, the probability of becoming non-natural while staying on level c is at most p.*

Proof. For either mutation operator, the probability of flipping a specified bit and no other bit is at least $1/(en)$. For standard bit mutation this follows from

$$\frac{1}{n}\left(1 - \frac{1}{n}\right)^{n-1} \ge \frac{1}{en},$$

and for one-bit mutation the probability is exactly $1/n$. A clean evaluation has probability $1 - p \ge 1/2$.

First consider a natural state with stored and true weight b. If $b < k$, flipping any zero-bit to one strictly improves the scalar value. There are $n - b \ge k - b \ge d(c)$ such bits. If $b > k$, flipping any one-bit to zero strictly improves it, and there are $b \ge b - k \ge d(c)$ choices. The clean single-bit events therefore have total probability at least s.

Next consider a non-natural state. Its true weight j is adjacent to its stored weight $b \in E_c$. Write $I_c = [\ell, r] \cap \mathbb{Z}$. We check all possible positions of j:

- If $j < \ell$, then $b = \ell$ and $j = \ell - 1$. Flipping a zero-bit reaches ℓ, and there are $n - \ell + 1 \ge k - \ell \ge d(c)$ choices. If $j > r$, the symmetric move uses $r + 1 \ge r - k \ge d(c)$ one-bits.
- If $\ell < j < r$, both adjacent weights lie in I_c. Every single-bit mutation therefore produces an acceptable weight.
- If j is a boundary of I_c, it must be the boundary opposite b. Indeed, $j \ne b$ and $|j - b| = 1$. The interval then has just two adjacent integer weights. Since it contains k and $b \ne k$, necessarily $d(c) = 1$. At least one available bit changes j back to $b \in I_c$.

Every offspring counted above, when evaluated cleanly, is accepted and natural. Its scalar value is either c or strictly smaller. This proves the second assertion. Finally, creating a non-natural retained state requires active evaluation noise, whose probability is p. □

Lemma 3 (Level tail and scalar-target time)**.** *Assume $p \le 1/(4en)$. Let L_c be the number of coordinate updates until a nonoptimal scalar level c is first left, starting from any admissible state on that level. With $\theta = 1/(16e)$,*

$$\mathbb{E}e^{\theta L_c/n} \le e^{2/d(c)}.$$

Let T_k be the number of coordinate updates until the stored weight first equals k, with the reference fixed at (n, n). Uniformly over admissible initial states,

$$\mathbb{E}e^{\theta T_k/n} \le e^{4H_n}, \qquad \mathbb{P}\left(T_k > \frac{n}{\theta}(4H_n + r)\right) \le e^{-r} \quad (r \ge 0). \tag{1}$$

Proof. The first step is to control the time on one level. Stop the process as soon as its scalar value becomes smaller than c, and use the potential

$$V_t = \begin{cases} 1, & \text{the process is still on level } c \text{ and its state is natural,} \\ 2, & \text{the process is still on level } c \text{ and its state is non-natural,} \\ 0, & \text{the process has left level } c. \end{cases}$$

The extra unit assigned to a non-natural state pays for repairing its stored evaluation. From $V_t = 1$, a strict improvement decreases the potential by one, whereas entering a non-natural state on the same level increases it by one. Thus lemma 2 gives expected decrease at least $s - p$. From $V_t = 2$, the potential cannot increase, and the cleanup event decreases it by at least one. Its expected decrease is therefore at least s.

Since $p \le 1/(4en) \le s/2$, both cases can be expressed as the same contraction:

$$\mathbb{E}[V_{t+1} \mid \mathcal{F}_t] \le (1 - q)V_t, \qquad q = \frac{s}{2} = \frac{d(c)}{4en}.$$

Iterating gives $\mathbb{E}V_t \le 2(1 - q)^t$. If the level has not yet been left, the potential is at least one. Consequently

$$\mathbb{P}(L_c > t) \le \mathbb{E}V_t \le 2(1 - q)^t.$$

This bound in particular proves that the level is left almost surely.

We use an exponential moment to combine the times spent on different levels without losing another logarithm. Put $\lambda = \theta/n$. Then $\lambda \le q/4$, and

$$e^\lambda - 1 \le 2\lambda, \qquad 1 - e^\lambda(1 - q) \ge q - 2\lambda \ge q/2.$$

The tail-sum identity for a nonnegative integer-valued random variable now gives

$$\begin{aligned} \mathbb{E}e^{\lambda L_c} &= 1 + (e^\lambda - 1)\sum_{t \ge 0} e^{\lambda t}\mathbb{P}(L_c > t) \\ &\le 1 + \frac{2(e^\lambda - 1)}{1 - e^\lambda(1 - q)} \le 1 + \frac{8\lambda}{q} = 1 + \frac{2}{d(c)} \le e^{2/d(c)}. \end{aligned}$$

It remains to combine the levels. The stored scalar value never increases, so a level that has been left cannot be revisited. List all nonoptimal scalar values in decreasing order and assign duration zero to any skipped level. At the entry to a visited level, the bound just proved holds conditionally on the entire entry history, because it is uniform over admissible states. Successive

conditioning therefore multiplies these bounds. Independence of level durations is not needed. Since at most two levels have each distance d, we obtain

$$\mathbb{E}e^{\theta T_k/n} \le \prod_{c>f_k(k)} e^{2/d(c)} \le \exp\left(4\sum_{d=1}^{n}\frac{1}{d}\right) = e^{4H_n}.$$

Markov's inequality gives the second bound in (1). □

The uniform tail is useful because the $n+1$ subproblems run together. Taking their maximum costs an additive $O(n\log n)$ term, rather than multiplying their expected times by $n+1$.

Proposition 4 (Phase I under one-bit noise)**.** *Under the assumptions of lemma 3, the Phase-I completion boundary S satisfies $\mathbb{E}S = O(n\log n)$ sweeps.*

Proof. First we bound the time until the reference becomes (n,n). The endpoint coordinate $k=n$ accepts precisely the nondecreasing observed weights, even while the reference moves. After the reference update, z_1 dominates both the candidate observation c and the stored observation b. Therefore its acceptance condition is

$$z_1 - c \le z_1 - b \quad\Longleftrightarrow\quad c \ge b.$$

Similarly, coordinate $k=0$ accepts precisely the nonincreasing observed weights. These two endpoint processes are thus exactly the scalar-target processes analyzed in lemma 3, with targets n and 0, respectively. Once both targets have been stored, their observations have made both reference components equal to n. The tail bound (1), followed by a union bound for these two coordinates, gives $O(n\log n)$ expected sweeps until the reference is ideal.

Now start at the first sweep boundary with reference (n,n), and condition on the full history at that boundary. Every coordinate is still admissible. For each k, let T_k denote its remaining target time in coordinate updates. Because every coordinate receives exactly one update per sweep, the number of additional sweeps needed for all targets is $\max_k T_k$. A stored target, once attained, cannot be lost.

Applying (1) conditionally to every coordinate and using a union bound gives, for $u \ge 0$,

$$\mathbb{P}\left(\max_k T_k > \frac{n}{\theta}(4H_n + \log(n+1) + u) \,\middle|\, \mathcal{F}_{\text{start}}\right) \le e^{-u}.$$

Here $\mathcal{F}_{\text{start}}$ denotes the history at the chosen boundary. Tail integration, with an immaterial rounding term for integer times, bounds the expected maximum by $O(n\log n)$. Adding the time needed to establish the reference proves the proposition. This argument requires no independence between the states at the starting boundary. □

4 A common repair potential for all four combinations

Correct stored weights do not necessarily mean correct true weights. Once Phase II starts, coordinate k always stores observation k, but its retained string may have a different true weight. We measure the remaining error by the expected time needed to repair that coordinate. This definition gives the same potential construction for all four mutation–noise combinations.

Let B and D be the weight transition matrices of one-bit mutation and standard bit mutation. Their entries are

$$B_{i,i+1} = \frac{n-i}{n}, \qquad B_{i,i-1} = \frac{i}{n},$$

with all other entries zero and nonexistent neighbours omitted, and

$$D_{i,j} = \sum_{\substack{0\le a\le i,\ 0\le u\le n-i\\ i-a+u=j}} \binom{i}{a}\binom{n-i}{u} n^{-(a+u)}(1-1/n)^{n-a-u}.$$

In this sum, a counts one-bits changed to zero and u counts zero-bits changed to one. The mutation matrix is $M \in \{B, D\}$. The observation matrix is

$$N = (1-p)I + pL, \qquad L \in \{B, D\},$$

so $N_{j,k}$ is the probability of observing weight k from a string of true weight j.

Both M and N are reversible with respect to

$$\mu_i = 2^{-n}\binom{n}{i}.$$

Indeed, on bit strings, each of the mutation and noise matrices is symmetric: a specified transition has the same probability as its reversal. Summing over pairs of strings with fixed weights gives

$$\mu_i M_{i,j} = \mu_j M_{j,i}, \qquad \mu_i N_{i,j} = \mu_j N_{j,i}.$$

Fix coordinate k in Phase II. An offspring is accepted exactly when its observed weight is k. Hence its true-weight chain has transition matrix

$$P_k(i,j) = M_{i,j}N_{j,k} \quad (i \neq j), \qquad P_k(i,i) = 1 - \sum_{j \neq i} M_{i,j}N_{j,k}.$$

The diagonal includes rejected offspring and accepted offspring of unchanged weight. Possible retained states form the set

$$\mathcal{S}_k = \{i \in \{0, \dots, n\} : N_{i,k} > 0\}.$$

For $0 < p < 1$, this is either $\{k-1, k, k+1\} \cap \{0, \dots, n\}$ under one-bit noise or the entire weight space under bitwise noise. In either case both mutation operators allow adjacent transitions, and those transitions have positive acceptance probability. Thus P_k is irreducible on $\mathcal{S}_k$.

Lemma 5 (Stationary law and exact repair identity)**.** *Assume $0 < p < 1$. The stationary distribution of the coordinate chain is*

$$\pi_k(i) = \frac{\mu_i N_{i,k}}{\mu_k} = N_{k,i}.$$

For this chain, define the repair potential

$$h_k(i) = \mathbb{E}_i \inf\{t \geq 0 : X_t = k\},$$

where $(X_t)_{t \geq 0}$ is the chain with transition matrix P_k, t counts coordinate updates, and $X_0 = i$. In particular $h_k(k) = 0$. The expected repair potential created by one update from a correct coordinate is

$$\lambda_k := \sum_j P_k(k,j)h_k(j) = \frac{1 - N_{k,k}}{N_{k,k}} \leq \frac{p}{1-p}. \tag{2}$$

Proof. Detailed balance for N gives $\mu_i N_{i,k}/\mu_k = N_{k,i}$. Thus the proposed probabilities sum to one. For $i \neq j$, detailed balance for M also gives

$$\begin{aligned}\pi_k(i)P_k(i,j) &= \frac{\mu_i M_{i,j} N_{i,k} N_{j,k}}{\mu_k} \\ &= \frac{\mu_j M_{j,i} N_{j,k} N_{i,k}}{\mu_k} = \pi_k(j)P_k(j,i).\end{aligned}$$

This proves reversibility and hence stationarity of π_k.

The repair times are finite because the chain is finite and irreducible. More explicitly, from each state choose a finite path to k with positive probability. The maximum path length and minimum path probability over this finite collection yield a geometric bound in blocks on the time to hit k.

First-step conditioning now gives

$$h_k(i) - \sum_j P_k(i,j)h_k(j) = 1 \quad (i \neq k).$$

At $i = k$, the same expression equals $-\lambda_k$, since $h_k(k) = 0$. Multiply these expressions by $\pi_k(i)$ and sum over all states. Stationarity makes the total left-hand side zero:

$$\begin{aligned} 0 &= \sum_i \pi_k(i)h_k(i) - \sum_j \left(\sum_i \pi_k(i)P_k(i,j) \right) h_k(j) \\ &= \sum_{i \neq k} \pi_k(i) - \pi_k(k)\lambda_k = 1 - \pi_k(k) - \pi_k(k)\lambda_k. \end{aligned}$$

Solving for λ_k and using $\pi_k(k) = N_{k,k} \geq 1 - p$ proves (2). □

The identity has a direct interpretation. An incorrect coordinate decreases its expected remaining repair time by one in expectation. A correct coordinate can become incorrect, but the expected repair time thereby created is at most $p/(1-p)$. Summing these coordinate potentials yields positive drift whenever the total noise contribution is sufficiently small.

Theorem 6 (Four-case Phase-II drift)**.** *Write $J_{k,t}$ for the true weight of coordinate k after t Phase-II sweeps. Define*

$$\mathcal{H}_t = \sum_{k=0}^{n} h_k(J_{k,t}), \qquad T_{\mathrm{II}} = \inf\{t \geq 0 : J_{k,t} = k \text{ for every } k\}.$$

For every Phase-II initial distribution and $0 < p < 1/(n+1)$,

$$\mathbb{E}T_{\mathrm{II}} \leq \frac{\mathbb{E}\mathcal{H}_0}{1 - np/(1-p)}.$$

Proof. The potential is nonnegative and vanishes exactly when every coordinate is truly correct. Suppose that $m \geq 1$ coordinates are incorrect at the start of a sweep. The first-step equations for h_k give expected decrease one for each incorrect coordinate. A correct coordinate instead has expected decrease $-\lambda_k$. Therefore

$$\begin{aligned} \mathbb{E}[\mathcal{H}_t - \mathcal{H}_{t+1} \mid \mathcal{F}_t] &= m - \sum_{k : J_{k,t} = k} \lambda_k \\ &\geq m - (n+1-m)\frac{p}{1-p} \geq 1 - \frac{np}{1-p} =: \delta > 0. \end{aligned}$$

Each coordinate receives one update per sweep. During Phase II its transition law is unaffected by the other coordinates, because the reference and stored weights are fixed. Thus this calculation remains valid even if the initial coordinate states are dependent.

We stop the analysis at the first simultaneous correct state; we do not assume that this state is absorbing in the original algorithm. For an integer $r \geq 1$, summing the conditional drift for the stopped process gives

$$\delta \sum_{t=0}^{r-1} \mathbb{P}(T_{\mathrm{II}} > t) \leq \mathbb{E}\mathcal{H}_0 - \mathbb{E}\mathcal{H}_{T_{\mathrm{II}} \wedge r} \leq \mathbb{E}\mathcal{H}_0.$$

Letting $r \to \infty$, the tail-sum formula and monotone convergence prove the bound. When $p = 0$, retained observations equal true weights, so Phase II has length zero under the stated initialization. □

To apply this common drift theorem, we must bound its initial potential. Under one-bit noise, such a bound holds for every admissible Phase-II state.

Proposition 7 (Uniform initialization for one-bit noise)**.** *For either mutation operator under one-bit noise,*

$$\mathcal{H}_0 \le \frac{2enH_{n+1}}{1-p}$$

for every admissible Phase-II configuration.

Proof. An incorrect retained weight in coordinate k is either $k-1$ or $k+1$. From $k-1$, exactly $n-k+1$ single-bit changes produce weight k. From $k+1$, exactly $k+1$ single-bit changes do so. Let

$$r_k = \min\{k+1, n-k+1\}.$$

For either mutation operator, the probability of a clean exact single-bit correction from any incorrect state is at least

$$q_k = \frac{(1-p)r_k}{en}.$$

If an update fails to repair the coordinate, its retained state is still in $\{k-1,k,k+1\}\cap\{0,\dots,n\}$. Conditional on still being incorrect, the same lower bound q_k therefore applies at the next update. The probability of remaining incorrect for t updates is at most $(1-q_k)^t$, and the tail-sum formula gives

$$h_k(i) \le \frac{1}{q_k} = \frac{en}{(1-p)r_k} \quad (i \ne k).$$

Each positive integer occurs at most twice among $r_0,\dots,r_n$. Since $h_k(k)=0$, summing over all coordinates gives

$$\mathcal{H}_0 \le \frac{en}{1-p}\sum_{k=0}^{n}\frac{1}{r_k} \le \frac{2enH_{n+1}}{1-p}.$$

□

Proof of the one-bit-noise part of theorem 1. By proposition 4, Phase I takes $O(n\log n)$ expected sweeps. At its completion boundary, proposition 7 bounds the repair potential by $O(n\log n)$ for every possible state. We may therefore apply theorem 6 conditionally on that boundary state. The denominator $1-np/(1-p)$ is bounded below by an absolute positive constant in the parameter range of theorem 1, so Phase II also takes $O(n\log n)$ expected sweeps. Multiplying the total by $n+1$ evaluations per sweep and adding the initialization cost proves the stated $O(n^2\log n)$ bound. □

5 A harmonic potential for stored-value progress

We next prepare the analysis under bitwise noise. The potential in this section measures progress in the stored value alone. Its drift is useful whenever the true and stored weights agree. The construction works for both mutation operators and both noise models.

Fix the ideal reference. If the current stored weight is b, all observations that would be accepted form the interval $I_{f_k(b)} = [\ell, r]\cap\mathbb{Z}$. A strict improvement shrinks this interval. We assign a cost to each weight that still lies between an interval boundary and the target k:

$$F_k(b) = n\sum_{w=\ell}^{k-1}\frac{1}{n-w} + n\sum_{w=k+1}^{r}\frac{1}{w}. \tag{3}$$

An empty sum is zero, so the formula also covers $k=0,n$. The summand $n/(n-w)$ is the reciprocal of the probability of a one-bit move from w to $w+1$. Similarly, n/w corresponds to a move from w to $w-1$. These reciprocals make the expected progress constant. Using the entire sublevel interval also handles accepted moves from one side of k to the other.

Lemma 8 (Harmonic progress)**.** *The potential F_k is nonnegative, vanishes exactly at $b = k$, and never increases under accepted updates with ideal reference. At a natural active state its expected decrease is at least*

$$a = (1-p)q_n, \qquad q_n = \begin{cases} 1, & \text{one-bit mutation,} \\ (1-1/n)^{n-1}, & \text{standard bit mutation.} \end{cases}$$

Moreover,

$$\sum_{k=0}^{n} \max_b F_k(b) \le 2n^2.$$

Proof. Smaller scalar values have nested, smaller sublevel intervals, proving monotonicity. If a natural stored weight $b < k$, it is the left endpoint of its sublevel interval. A clean single-bit step to $b+1$ strictly improves the scalar and removes at least the summand $n/(n-b)$. Its probability is $(1-p)q_n(n-b)/n$, so it contributes at least a to expected decrease. The case $b > k$ uses the summand n/b and probability $(1-p)q_n b/n$. Every other update has nonnegative decrease by nested sublevel sets.

For each k,

$$F_k(b) \le n(H_n - H_{n-k}) + n(H_n - H_k).$$

Summing and using $\sum_{k=0}^{n} H_k = (n+1)H_n - n$ proves $2n^2$. □

Thus, after the reference becomes ideal, the expected total number of natural active updates is at most $2n^2/a$. To obtain this bound, sum the conditional decreases over individual updates and discard the nonnegative final potential. Non-natural updates do not increase F_k, but they need not decrease it. We will control their number with a separate repair potential.

The estimate above uses the fixed ideal reference. While the reference moves, a sublevel interval can expand and the potential can increase. The next section handles that initial period separately.

6 Standard bit mutation with activated bitwise noise

Here $M = D$ and $N = (1-p)I + pD$. Throughout this section assume $0 < p \le 1/(16e(n+1))$, unless a lemma states a larger range. Write

$$c_p = \frac{p}{1-2p}, \qquad a = (1-p)(1-1/n)^{n-1} \ge \frac{1}{2e}.$$

The key distinction is between updates that start in a natural state and those that start with an observation error. After the reference becomes ideal, we will bound the remaining number of sweeps by

$$\max_k \{\text{natural active updates of coordinate } k\} + \sum_k \{\text{non-natural active updates of coordinate } k\}.$$

A fitness-level argument controls the maximum. Additive drift of a repair potential controls the sum. To make that second bound useful, we first estimate how much expected repair time one noisy evaluation can create.

6.1 The expected repair cost of one observation

For any stored weight b, consider the auxiliary chain that accepts an offspring exactly when its observed weight is b. Let $h_b(i)$ be its expected time to reach true weight b, starting from true

weight i. This is the repair chain of section 4, with b in place of k. It measures correction of the observation error even when b is not the subproblem's target.

The probability that the next observed weight equals b is $(1-p)D_{i,b} + p(D^2)_{i,b}$. The first term corresponds to a clean evaluation. The second corresponds to mutation followed by active bitwise noise. Both can lead to acceptance, which is why both enter the repair bound below.

Lemma 9 (Repair bound using two composed mutations)**.** *For $0 < p < 1/2$, define*

$$A_b(i) = (1-p)D_{i,b} + p(D^2)_{i,b}, \qquad C_p = \frac{1-p}{1-2p}.$$

The expected repair time in the chain pinned at observation b satisfies

$$h_b(i) \le \frac{C_p}{A_b(i)} \quad (i \neq b), \qquad N_{i,b}h_b(i) \le c_p \quad \text{for all } i. \tag{4}$$

Here D^2 is matrix multiplication, the weight kernel of two consecutive independent standard mutations.

Proof. We construct a potential whose expected decrease is at least one until repair. Set $v(b) = 0$ and $v(i) = C_p/A_b(i)$ for $i \neq b$. The chain accepts only observation b. For $i \neq b$, its expected decrease is

$$\begin{aligned} v(i) - \sum_j P_b(i,j)v(j) &= A_b(i)v(i) - p\sum_{j\neq b} D_{i,j}D_{j,b}v(j) \\ &= C_p\left(1 - p\sum_{j\neq b}\frac{D_{i,j}D_{j,b}}{A_b(j)}\right) \\ &\ge C_p\left(1 - \frac{p}{1-p}\right) = 1. \end{aligned}$$

In the first line, $A_b(i)v(i)$ counts all accepted observations. The sum subtracts the potential left after accepted offspring of true weight $j \neq b$. Accepted offspring with unchanged true weight occur on both sides and cancel. The inequality uses $A_b(j) \ge (1-p)D_{j,b}$ and $\sum_j D_{i,j} = 1$.

Let τ be the first hit of b. Summing the drift up to $\tau \wedge r$ gives $\mathbb{E}(\tau \wedge r) \le v(i)$. Letting $r \to \infty$ proves $h_b(i) \le v(i)$. Finally, if $i \neq b$ then $N_{i,b} = pD_{i,b}$, so

$$N_{i,b}h_b(i) \le \frac{pD_{i,b}C_p}{(1-p)D_{i,b}} = c_p.$$

For $i = b$ the left side is zero. □

The product $N_{i,b}h_b(i)$ has a simple meaning: it is the contribution of observation b to the expected repair time created by evaluating a true weight i. Summing the bound c_p over $b \neq i$ gives nc_p. The next lemma improves that sum by showing that the contribution decreases geometrically with the error distance $|i-b|$.

Lemma 10 (Total repair cost of an observation)**.** *For $0 < p \le 1/4$, uniformly in the true weight i,*

$$\sum_{b=0}^{n} N_{i,b}h_b(i) \le \eta, \qquad \eta = \min\left\{nc_p,\ 24p\log\frac{e}{p}\right\}. \tag{5}$$

Proof. Step 1: compare one mutation with two. We claim that

$$(D^2)_{i,b} \ge \frac{1}{4}(3/2)^{|i-b|}D_{i,b}. \tag{6}$$

Indeed, composing two independent bit mutations with per-bit rate $q = 1/n$ yields independent bit flips with rate $q' = 2q(1-q)$. For strings x, y at Hamming distance r, the ratio of the composed transition probability to the original one is

$$\left(\frac{1-q'}{1-q}\right)^n \left(\frac{q'(1-q)}{q(1-q')}\right)^r.$$

For $n \geq 3$ its first factor is at least $(1-1/n)^n \geq 1/4$, and the second base is at least $3/2$. For the latter assertion, substitute $q \leq 1/3$ and check $1/2 - q - q^2 \geq 0$. Sum over strings of weight b, using $r \geq |i-b|$, to obtain (6). For $n = 2$, $D^2 = D$ and the displayed inequality holds directly.

Step 2: bound the contribution at each distance. For $d = |i-b| \geq 1$, lemma 9 gives

$$N_{i,b}h_b(i) \leq \frac{C_p p}{(1-p) + (p/4)(3/2)^d} \leq \min\{2p,\ 6(3/2)^{-d}\}.$$

Here $C_p \leq 3/2$ for $p \leq 1/4$. Thus every contribution is at most $2p$, and large errors have a stronger geometric bound.

Step 3: sum over distances. At most two observed weights have each distance d. Put $L = \lceil \log(3/p)/\log(3/2) \rceil$. Use the $2p$ bound up to L and the geometric bound thereafter:

$$\sum_b N_{i,b}h_b(i) \leq 4pL + 12\sum_{d>L}(3/2)^{-d} \leq 4pL + 8p \leq 24p\log(e/p).$$

The last numerical inequality follows from $4/\log(3/2) < 10$ and $4\log 3/\log(3/2) + 12 < 24$. The alternative bound nc_p follows by summing (4) over the n weights $b \neq i$. □

6.2 Repair while the reference moves

The preceding estimates concern an auxiliary chain with a fixed observation. The actual algorithm may accept a different observation and may update its reference. The following comparison transfers the estimate to that algorithm.

Lemma 11 (Repair potential while the reference moves)**.** *For one coordinate put $R(j,b) = h_b(j)$. On every update, for any current reference, its conditional expected increase is at most*

$$\begin{cases} -1+\eta, & j \neq b, \\ \eta, & j = b. \end{cases}$$

Its expectation immediately after an initial noisy evaluation is at most η, for any true initial string.

Proof. If $j = b$, the old potential is zero. The expected new potential is bounded by summing over all possible accepted observations. By lemma 10 this is at most $\sum_\ell D_{j,\ell}\eta = \eta$.

If $j \neq b$, compare with the auxiliary chain accepting only observation b. The actual algorithm always accepts observation b, even after a reference update, because both compared scalar values are equal. In the auxiliary chain the expected increase of h_b is -1, by its first-step equation. An additional acceptance with observation $c \neq b$ replaces the auxiliary chain's nonnegative potential by $h_c(\ell)$, where ℓ is the true offspring weight. Its extra contribution is therefore at most $h_c(\ell)$. Summing gives

$$\sum_\ell D_{j,\ell}\sum_{c\neq b} N_{\ell,c}h_c(\ell) \leq \eta.$$

Finally the initial bound is exactly (5). □

Lemma 12 (Reference time and accumulated repair potential)**.** *For the reference time U defined in section 2,*

$$\mathbb{E}U = O(n\log n), \qquad \mathbb{E}\sum_{k=0}^{n} h_{b_{k,U}}(j_{k,U}) \leq (n+1)\eta(1+\mathbb{E}U) = O(n^2\eta\log n).$$

Proof. Reaching the ideal reference. Since $c_p \le 2p$ in our parameter range,

$$\eta \le nc_p \le \frac{1}{8e}.$$

At endpoint zero use $F_0(b) = nH_b$, and at endpoint n use $F_n(b) = nH_{n-b}$. Their observation selection is monotone from the beginning of the run, regardless of the reference. By lemmas 8 and 11, the potential $F_k + R$, set to zero on first storing the endpoint target, has expected decrease at least $a - \eta \ge 3/(8e)$ until stopping. At a natural active state this follows from drift a in F_k. At a non-natural state it follows from drift at least $1 - \eta$ in R, while F_k cannot increase. Killing on success can only improve these inequalities.

The initial expectation is at most $nH_n + \eta$. Additive drift therefore bounds each endpoint's target time by $O(n \log n)$ coordinate updates. The reference is ideal no later than both endpoints have stored their targets, proving the bound on U.

Bounding the repair potential at that time. On every coordinate update, the expected increase of R is at most η. Hence its total expected increase per sweep is at most $(n+1)\eta$. Sum this bound up to $U \wedge r$. The initial expected sum is at most $(n+1)\eta$. Nonnegativity and Fatou's lemma as $r \to \infty$ give the second bound. □

6.3 Finishing Phase I after the reference is ideal

We now reset the local time to zero at a sweep boundary with ideal reference. For each coordinate, count natural and non-natural updates separately until its stored target is reached. The maximum natural count controls progress across the parallel subproblems. The sum of non-natural counts is an upper bound on all additional delay.

Lemma 13 (Counting natural updates separately from repair)**.** *Start at any sweep boundary with ideal reference. For coordinate k, let τ_k be the number of its updates until its stored weight first equals k. Count updates up to and including the successful one according to their starting states. Let A_k count natural active updates, and B_k non-natural active updates. Thus $\tau_k = A_k + B_k$, allowing infinite counts until finiteness has been proved. If the stored target has already been reached, both counts are zero. Define*

$$R_0^{\mathrm{tot}} = \sum_{k=0}^{n} h_{b_{k,0}}(j_{k,0}).$$

Then

$$\mathbb{E} \max_k A_k = O(n \log n), \qquad \sum_k \mathbb{E} A_k \le 2n^2/a, \tag{7}$$

$$(1-\eta) \sum_k \mathbb{E} B_k \le \mathbb{E} R_0^{\mathrm{tot}} + \eta \sum_k \mathbb{E} A_k. \tag{8}$$

Consequently, the expected time to reach all stored targets is at most $\mathbb{E} \max_k A_k + \sum_k \mathbb{E} B_k$ sweeps, whenever $\mathbb{E} R_0^{\mathrm{tot}} < \infty$.

Proof. Step 1: the natural updates of one coordinate. Let $A_{k,c}$ count the natural updates of coordinate k on a nonoptimal scalar level c. At each such update, a clean exact single-bit mutation toward k leaves the level with probability at least

$$s_c = \frac{d(c)}{2en}.$$

This is the natural-state argument of lemma 2, which uses only the probability of a clean evaluation.

Expose successive natural updates on this level, ignoring intervening non-natural updates. Each exposed update has conditional probability at least s_c of leaving the level permanently. Hence $A_{k,c}$ is stochastically dominated by a geometric random variable of parameter s_c on $\{1, 2, \ldots\}$. If a non-natural update leaves the level, or if the process never returns to a natural state, the count is only smaller.

For $\theta = 1/(8e)$ and $\lambda = \theta/n \le s_c/4$, the geometric moment bound gives

$$\mathbb{E}e^{\lambda A_{k,c}} \le 1 + \frac{e^\lambda - 1}{1 - (1 - s_c)e^\lambda} \le 1 + \frac{4\lambda}{s_c} \le e^{1/d(c)}.$$

This bound is uniform conditional on the history at entry to the level. As in lemma 3, multiply over the decreasing list of possible scalar levels, assigning zero counts to skipped levels. Formally, apply this argument to counts truncated at a finite local time, and then pass to the limit by monotone convergence. At most two levels have any given $d(c)$, so

$$\mathbb{E}e^{\theta A_k/n} \le e^{2H_n}.$$

Markov's inequality and a union bound yield

$$\mathbb{P}\left(\max_k A_k > \frac{n}{\theta}(2H_n + \log(n+1) + u)\right) \le e^{-u} \quad (u \ge 0).$$

Integrating this tail proves $\mathbb{E}\max_k A_k = O(n \log n)$. We have bounded counts of natural updates without yet assuming finite target times.

Step 2: the total number of natural updates. By lemma 8, every natural active update decreases F_k by at least a in expectation, and all other updates have nonnegative decrease. Sum this drift up to a finite time, then use nonnegativity and monotone convergence. Since the total initial potential is at most $2n^2$, this gives

$$a\sum_k \mathbb{E}A_k \le 2n^2.$$

Step 3: the non-natural updates. Let $R_{k,t}$ equal $h_b(j)$ after the tth local update of coordinate k, until its stored target is first reached. Set it to zero from that time onwards. Let $A_k(r)$ and $B_k(r)$ count the two types of updates among its first $\tau_k \wedge r$ updates. By lemma 11, a natural update increases the expected repair potential by at most η, while a non-natural update decreases it by at least $1 - \eta$. Therefore

$$0 \le \mathbb{E}R_{k,\tau_k\wedge r} \le \mathbb{E}R_{k,0} + \eta\mathbb{E}A_k(r) - (1-\eta)\mathbb{E}B_k(r).$$

Sum over k and let $r \to \infty$. Monotone convergence and $\sum_k R_{k,0} \le R_0^{\mathrm{tot}}$ prove (8). When the initial expected repair potential is finite, both counts have finite expectation. Every $\tau_k = A_k + B_k$ is then almost surely finite.

Finally, each coordinate receives one update per sweep, and a stored target stays fixed. The remaining sweep count is $\max_k \tau_k$, which satisfies

$$\max_k \tau_k \le \max_k A_k + \sum_k B_k.$$

□

6.4 Entering Phase II and combining the bounds

The preceding lemmas bound Phase I. To apply the common Phase-II theorem, we still need its initial repair potential. The next argument tracks only coordinates whose stored observation is already their target. It remains valid while the reference moves.

Lemma 14 (Initializing the common Phase-II potential)**.** *Suppose the initial true strings are evaluated independently by the stated noise model, with any initial reference dominating those observations. At the Phase-I completion sweep S,*

$$\mathbb{E}\mathcal{H}_S \le (n+1)c_p(1+\mathbb{E}S), \qquad \mathcal{H}_S = \sum_k h_k(j_{k,S}).$$

Proof. Throughout the run, define

$$Q_k = \mathbf{1}\{b_k = k\}h_k(j_k).$$

If $b_k \neq k$, then $Q_k = 0$. It can increase only when an observation k is accepted. By lemma 9, the expected increase is at most

$$\sum_\ell D_{j_k,\ell}N_{\ell,k}h_k(\ell) \le c_p.$$

If $b_k = k$, every new observation k is accepted. Relative to the chain accepting only observation k, accepting any other observation sets Q_k to zero and can only reduce it. Its expected increase is therefore at most -1 when $j_k \neq k$, and at most $\lambda_k \le p/(1-p) \le c_p$ when $j_k = k$.

Initially $\mathbb{E}Q_k \le c_p$, again by lemma 9. Each sweep increases the expected sum by at most $(n+1)c_p$. Summing up to $S \wedge r$, then applying nonnegativity and Fatou's lemma, gives the bound. At time S, every stored observation equals its target, so $\sum_k Q_k = \mathcal{H}_S$. Here $\mathcal{H}_S$ uses the global time S and is the initial potential $\mathcal{H}_0$ in the local Phase-II clock. □

Theorem 15 (Bitwise mutation and bitwise noise)**.** *For the cyclic $T = 1$ algorithm, with initial strings evaluated independently and the reference initialized from their observations, suppose $0 < p \le 1/(16e(n+1))$. Then*

$$\mathbb{E}[\textit{evaluations until exact covering}] = O\left(n^2 \log n \left[1 + np\log\frac{e}{p}\right]\right). \tag{9}$$

The implied constant is independent of n and p. In particular, the bound is $O(n^2\log^2 n)$ throughout this range and $O(n^2 \log n)$ if $p = O(1/(n\log n))$. At $p = 0$, it is $O(n^2\log n)$.

Proof. Phase I. First spend U sweeps reaching the ideal reference. Apply lemma 13 to the state at that boundary. The reference lemma bounds its expected repair potential, and the natural-count lemma bounds both types of subsequent updates. Together they give

$$\begin{aligned}\mathbb{E}S &\le O(n\log n) + \frac{O(n^2\eta\log n) + \eta\, O(n^2)}{1-\eta}\\ &= O(n\log n + n^2\eta\log n).\end{aligned}$$

The first term includes $\mathbb{E}U$ and the expected maximum natural count. The terms in the numerator bound, respectively, the repair potential at U and the repair potential created by subsequent natural updates. We use $\eta \le 1/(8e)$ to bound the denominator away from zero.

Phase II. By lemma 14 and theorem 6, the remaining expected number of sweeps is at most

$$\frac{(n+1)c_p(1+\mathbb{E}S)}{1-np/(1-p)} = O(1+\mathbb{E}S).$$

The factor $(n+1)c_p$ is bounded, and the denominator is bounded below by an absolute positive constant. Hence both phases together take $O(n\log n + n^2\eta\log n)$ expected sweeps. Multiplying by $n+1$, adding initialization, and substituting $\eta \le 24p\log(e/p)$ proves (9).

Finally, $p\log(e/p)$ is increasing on $(0,1)$. Thus $p \le C/n$ gives $np\log(e/p) = O(\log n)$, while $p \le C/(n\log n)$ gives $np\log(e/p) = O(1)$, for any fixed C. At $p = 0$ the earlier noiseless case applies. □

6.5 Where the remaining logarithm enters

The accumulated repair potential at time U is the source of the remaining logarithm. To see this, first consider supplying the ideal reference at initialization.

Corollary 16 (Ideal reference supplied initially)**.** *If the reference is initialized to (n, n) and each true initial string is evaluated once by the stated bitwise-noise model, the cyclic $T = 1$ algorithm with standard bit mutation has expected exact-covering cost $O(n^2 \log n)$ for all $p \le 1/(16e(n+1))$.*

Proof. For $p > 0$, the initial expected total repair potential is at most $(n+1)\eta$. Lemma 13 therefore gives

$$\mathbb{E}S = O(n \log n + \eta n^2) = O(n \log n),$$

since $\eta = O(\log n/n)$. Lemma 14 and theorem 6 give the same order for Phase II. Multiplication by $n+1$ proves the claim. At $p = 0$, use the earlier noiseless case. □

This corollary changes the initial reference. It is not used to prove the bound for the original initialization.

For the original reference rule, our estimate at time U is

$$\mathbb{E}\sum_k h_{b_{k,U}}(j_{k,U}) \le (n+1)\eta(1 + \mathbb{E}U) = O(n \log^2 n).$$

An improvement to

$$\mathbb{E}\sum_k h_{b_{k,U}}(j_{k,U}) = O(n \log n) \tag{10}$$

would give $O(n^2 \log n)$ evaluations throughout the stated noise range: substitute it into (8), then apply lemma 14 and theorem 6. This estimate remains open. The harmonic potential alone does not prove it, because moving the reference can change the minimizing weight and enlarge a sublevel interval.

7 Why one-bit mutation with bitwise noise is different

Standard bit mutation can correct an observation error of size two in a single clean update. One-bit mutation cannot. The following lower bound shows that this difference can affect the order of the expected runtime, even when noise is rare.

Theorem 17 (A lower bound for one-bit mutation and activated bitwise noise)**.** *Consider MOEA/D as defined above, with neighbourhood size one, one-bit mutation, activated bitwise noise of probability $p = n^{-4}$, independent uniform initialization, and the reference updated from every observation. For all sufficiently large n divisible by 256,*

$$\mathbb{E}T_{\text{exact}} = \Omega(n^{3/2}) \quad \textit{in sweeps},$$

and hence the expected evaluation cost is $\Omega(n^{5/2})$. Consequently, for every fixed $\alpha > 0$, an $O(n^2 \log n)$ exact-covering bound cannot hold uniformly over $p \le \alpha/n$.

Proof idea. An evaluation that flips two one-bits can store weight $j-2$ for a true weight j. If both weights lie to the right of the scalar minimizer, this observation is accepted. A clean one-bit mutation then produces weight $j-1$ or $j+1$. Both exceed the stored value $b = j-2$ on the increasing scalar branch, so both are rejected. This coordinate must wait for another active-noise evaluation. We identify $\Theta(n^{3/2})$ early updates at which this can happen. For each update, require that it is the only active-noise evaluation through a specified initial period. These events are disjoint. Each has probability $\Omega(p)$ and forces a subsequent expected waiting

time of at least $1/p$, so their contributions add. The reference estimates in the appendix ensure that reference motion cannot release the incorrect state during this wait.

The full proof appears in appendix A. The lower bound uses the original uniform initialization and moving reference. It concerns exact covering, and does not establish the same bound for front coverage with arbitrary assignment.

8 Discussion

The results in function evaluations are summarized below. All upper bounds assume $p \leq 1/(16e(n+1))$ and use the original cyclic schedule with $T = 1$.

Mutation	Observation noise	Exact-covering bound
One-bit	One-bit	$O(n^2 \log n)$ upper bound.
Standard bit	One-bit	$O(n^2 \log n)$ upper bound.
Standard bit	Activated bitwise	$O(n^2 \log^2 n)$ upper bound; $O(n^2 \log n)$ if $p = O(1/(n \log n))$.
One-bit	Activated bitwise	$\Omega(n^{5/2})$ lower bound at $p = n^{-4}$, for all sufficiently large n divisible by 256.

The common repair potential explains how all coordinates can become truly correct at the same time. It decreases by one in expectation at each incorrect coordinate, while a correct coordinate creates at most $p/(1-p)$ expected repair cost. Completing a runtime proof also requires control of the potential at the end of Phase I. Under one-bit noise this control is uniform over all possible states. Under bitwise noise we instead bound the expected repair cost created along the run.

Two questions remain. For standard bit mutation with bitwise noise, can the estimate at reference freezing be improved to (10), giving $O(n^2 \log n)$ throughout the stated noise range? For one-bit mutation with bitwise noise, what is the runtime for front coverage when weights may appear in arbitrary subproblems? The lower bound at $p = n^{-4}$ also leaves open the exact-covering order under the separate restriction $p = \Theta(1/n)$.

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A Proof of the one-bit-mutation lower bound

Proof of theorem 17. Throughout the proof let

$$\mu = \frac{n}{2}, \qquad s = \frac{1}{4}\sqrt{n \log n}, \qquad m = \lfloor \sqrt{n}/16 \rfloor, \qquad H = 8n, \qquad K = \{1, \dots, n/64\}.$$

Write $\overline{w}$ and $\underline{w}$ for the largest and smallest weights observed so far. Thus $z_1 = \overline{w}$ and $z_2 = n - \underline{w}$. After the reference update, both observations being compared lie in $[\underline{w}, \overline{w}]$. The scalar minimizer on this interval is

$$a_k = \frac{k}{n}\overline{w} + \left(1 - \frac{k}{n}\right)\underline{w}. \tag{11}$$

The scalar objective is strictly increasing to the right of a_k for all $k \in K$.

Step 1: an initial configuration and endpoint progress. Let $W_0, \dots, W_n$ be the initial true weights. Define $\mathcal{I}$ by

$$\max_i W_i \le \mu + 4s, \qquad \min_i W_i \le \mu - s,$$
$$W_0 \le \mu + \sqrt{n}, \qquad \#\{k \in K : |W_k - \mu| \le \sqrt{n}\} \ge n/256.$$

We verify below that $\Pr(\mathcal{I}) \ge 1/2$ for sufficiently large n. Fix initial weights satisfying $\mathcal{I}$. Call a coordinate $k \in K$ *typical* if $|W_k - \mu| \le \sqrt{n}$. There are at least $n/256$ such coordinates.

We also need a bound on endpoint zero. Consider its hypothetical trajectory with clean observations, driven by its mutation randomness. Let $W_0(t)$ be its weight after t updates. This is one-bit ZeroMax: a decrement is accepted and an increment is rejected, independently of the reference. Until its weight is at most $n/4$, each update decreases it with probability at least $1/4$.

Couple the decrements with independent Bernoulli$(1/4)$ variables. A Chernoff bound shows that, at any integer time t, the probability of fewer than $t/8$ decrements while still above $n/4$ is at most $e^{-t/32}$. Consequently there is an event $\mathcal{A}$, depending only on endpoint mutation randomness, with probability $1 - O(e^{-s/4})$, such that for every integer $t \in [\lceil 8s \rceil, H]$, either the endpoint has already reached $n/4$, or

$$W_0(t) \le W_0(0) - t/8. \tag{12}$$

The probability estimate follows by summing $e^{-t/32}$ over these times and is uniform in the initial weights. On $\mathcal{A}$, the endpoint must reach $n/4$ by time $H = 8n$: otherwise (12) would make its weight negative. The noise events used next leave all endpoint evaluations clean, so this hypothetical trajectory will be the actual one.

Step 2: one erroneous observation. Consider an update ℓ of a typical coordinate during the first m sweeps. Let $\mathcal{E}_\ell$ be the event that all evaluations through sweep H, including initialization, have inactive noise except at ℓ. At that update, noise is active and flips exactly two one-bits of the offspring and no other bits. The events $\mathcal{E}_\ell$ for different updates are disjoint.

Before ℓ, every retained observation is exact. The true weight can change by at most one in each coordinate update. Hence the true offspring weight j at ℓ satisfies $|j - \mu| \le 2\sqrt{n}$, for sufficiently large n. The specified noise produces observed weight

$$b = j - 2 \ge \mu - 3\sqrt{n}.$$

Conditional on active noise, its probability is

$$\binom{j}{2} n^{-2}(1 - 1/n)^{n-2} \ge c_0$$

for an absolute constant $c_0 > 0$, since $j \geq n/3$ for large n. There are $R = (H+1)(n+1)$ evaluations through sweep H, including initialization. It follows that, for the fixed initial weights,

$$\Pr(\mathcal{E}_\ell \mid \mathcal{A}) \geq c_0 p(1-p)^{R-1} \geq c_0 p/2. \tag{13}$$

To justify the conditioning, first prescribe the single active-noise position ℓ. The bound on j then holds for every preceding mutation history, even after conditioning on $\mathcal{A}$. The remaining flip randomness at ℓ gives the factor c_0. Finally, $Rp = O(n^{-2})$, which gives the last inequality.

Step 3: the scalar minimizer stays below b. We work on $\mathcal{A} \cap \mathcal{E}_\ell$. During sweep $t+1$, a retained weight and its offspring differ from their initial weight by at most $t+1$. The sole noisy observation is smaller than its offspring weight. Since initialization is clean, the reference therefore satisfies

$$\overline{w} \leq \mu + 4s + t + 1, \qquad \underline{w} \leq \mu - s.$$

We bound (11) in three time ranges. Since $k/n \leq 1/64$ and $\overline{w} \geq \underline{w}$, replacing k/n by $1/64$ gives an upper bound.

For $0 \leq t \leq 8s+1$,

$$a_k \leq \frac{\mu + 4s + t + 1}{64} + \frac{63(\mu - s)}{64} \leq \mu - \frac{51s}{64} + \frac{2}{64} \leq \mu - \frac{s}{2}. \tag{14}$$

For $t \geq \lceil 8s \rceil$, while endpoint zero has not yet reached $n/4$, we additionally have $\underline{w} \leq W_0(t)$. Here t counts completed sweeps, so the endpoint has received at least t updates. Using (12) and $\sqrt{n} \leq s/8$, valid for sufficiently large n, gives

$$\begin{aligned} a_k &\leq \frac{\mu + 4s + t + 1}{64} + \frac{63(\mu + s/8 - t/8)}{64} \\ &= \mu + \frac{95s}{512} - \frac{55t}{512} + \frac{1}{64} \leq \mu - \frac{s}{2}. \end{aligned} \tag{15}$$

Finally, after the endpoint reaches $n/4$, monotonicity of the reference gives $\underline{w} \leq n/4$ permanently. Since $\overline{w} \leq n$ always,

$$a_k \leq \frac{k}{n} n + \left(1 - \frac{k}{n}\right) \frac{n}{4} \leq \frac{67n}{256}. \tag{16}$$

This last bound remains valid after sweep H, whatever the other coordinates subsequently do. Each of the three bounds is smaller than $b \geq \mu - 3\sqrt{n}$ for sufficiently large n. Thus $a_k < b$ at the noisy update and thereafter until at least sweep H; the permanent bound then continues to apply.

Step 4: acceptance and the ensuing waiting time. At update ℓ, the parent has exact stored weight i, and one-bit mutation gives $j = i \pm 1$. The new observed weight is $b = j - 2 < i$. By Step 3, both b and i lie to the right of a_k, where the scalar objective is strictly increasing. The offspring is therefore accepted. It has true weight j and stored weight $b = j - 2$.

Every subsequent clean one-bit mutation has weight $j-1$ or $j+1$. Both exceed b, so both are rejected while $a_k < b$. Step 3 therefore shows that this state persists through sweep H and cannot subsequently change at a clean evaluation. At time H, its true weight still satisfies $j \geq \mu - 2\sqrt{n} > k$. The first future active-noise evaluation of this coordinate occurs after a geometric number of updates of mean $1/p$, independently of the events through H. Since there is one update per sweep, exact covering cannot occur during this waiting time.

There was also no earlier exact-covering visit. Before ℓ, the same coordinate has weight at least $\mu - \sqrt{n} - m > k$; after ℓ it remains at $j > k$. Consequently, conditionally on the fixed initial weights,

$$\mathbb{E}[T_{\text{exact}} \mid \mathcal{A}, \mathcal{E}_\ell] \geq 1/p. \tag{17}$$

An active-noise evaluation need not repair the coordinate. We use only that repair cannot occur before it.

Step 5: add the disjoint contributions. There are at least $(n/256)m$ eligible updates ℓ. Combining (13) and (17), and summing over their disjoint events, gives a lower bound proportional to

$$(n/256)m\,p\,\frac{1}{p}$$

for the conditional expected runtime on $\mathcal{A}$. Finally, $\Pr(\mathcal{I}) \geq 1/2$ and $\Pr(\mathcal{A} \mid \text{initial weights}) = 1 - o(1)$ uniformly. Averaging therefore gives

$$\mathbb{E}T_{\text{exact}} \geq c(n/256)m = \Omega(n^{3/2})$$

for an absolute constant $c > 0$. Multiplying by $n + 1$ converts sweeps to evaluations. □

A.1 Probability of the initial event

We check the four conditions separately. For $W \sim \mathrm{Bin}(n, 1/2)$, Hoeffding's bound and a union bound give

$$\Pr(\max_i W_i > \mu + 4s) \leq (n+1)e^{-2(4s)^2/n} = O(1/n).$$

For the lower extreme, it is enough to bound one binomial mass. Put $r = \lceil s \rceil$. Stirling's inequalities give $2^{-n}\binom{n}{n/2} \geq c/\sqrt{n}$. Moreover,

$$\frac{\binom{n}{n/2-r}}{\binom{n}{n/2}} = \prod_{u=1}^{r}\left(1 - \frac{2u-1}{n/2+u}\right) \geq \exp\left(-\frac{4r(r+1)}{n}\right).$$

Indeed, $r = o(n)$, so each factor permits the estimate $\log(1-x) \geq -2x$, and $(2u-1)/(n/2+u) \leq 4u/n$. Thus $\Pr(W \leq \mu - s) \geq cn^{-3/4-o(1)}$. Independence of the initial weights now implies

$$\Pr(\min_i W_i > \mu - s) \leq \left(1 - cn^{-3/4-o(1)}\right)^{n+1} = o(1).$$

Next, Chebyshev's inequality gives $\Pr(|W - \mu| > \sqrt{n}) \leq 1/4$. The independent indicators of typical coordinates in K have success probability at least $3/4$. A Chernoff bound therefore shows that at least $|K|/2 = n/128 \geq n/256$ are typical, except with probability e^{-cn}. Finally, the same Chebyshev bound gives $\Pr(W_0 \leq \mu + \sqrt{n}) \geq 3/4$. The first three conditions just checked hold with probability $1 - o(1)$, while this last condition has probability at least $3/4$. A union bound therefore yields $\Pr(\mathcal{I}) \geq 1/2$ for sufficiently large n.